

Puzzles to Puzzle You (Remastered)

Welcome to the eighth iteration of *Puzzles to Puzzle You (Remastered)* by Rani Rachavelpula, in which I will discuss the previous puzzle (Mutilated Chessboard Problem) and pose the next one.

As noted in the previous newsletter, the mutilated chessboard problem is a tiling puzzle posed by Max Black in 1946. I, however, first heard of the problem in 2019 in a meeting with the great Haim Gaifman, who posed this question to me as part of a pop quiz testing whether I had what it takes to study logic. As my childhood friends can attest, I've always been a pressure player. Thankfully, I got it correct and avoided embarrassment.

Puzzle #7: Mutilated Chessboard Problem

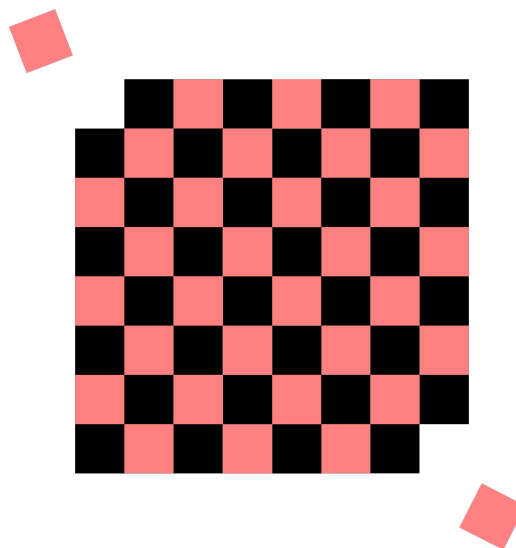
Consider a standard 8×8 chessboard with 64 squares colored alternately black and white. Suppose you have 32 dominoes, each of which is half black and half white. The board is initially covered completely by these 32 dominoes, where each domino covers exactly two adjacent squares such that the black half of the domino covers a black square and the white half covers a white square.

1. Now, suppose you remove the two opposite corner squares from the chessboard. Is it still possible to cover the remaining 62 squares exactly with 31 dominoes, each covering one black and one white square?

No

Shoutout (My Students) My summer was spent prepping and teaching Symbolic Logic. Somehow I was tasked with teaching an exceptionally talented group of students. I was especially delighted that my students shared my love for the Königsberg bridge problem, which I adapted in the final lecture to tease the beauty of model theory.

Solution If you did not get this puzzle, you will be kicking yourself.



Mutilated Chessboard Problem

Notice that removing opposite corners involves removing two pieces of the same color. This leaves 30 squares of one color and 32 of the other. However, each domino is half white, half black. Therefore, any tiling of dominoes must have an equal numbers of white and black squares. It follows by contrapositive that it is not possible to tile the mutilated board. $\square$

Puzzle #8: Knights and Knaves

Here is another puzzle from my logic class: On the island of Knights and Knaves, everyone is either a Knave or a Knight. *Knights always tell the truth. Knaves always lie.* Given the following statements, determine for each islander if they are a Knight or a Knave.

- Piggy says: Ralph always tells the truth.
- Ralph says: Jack is a Knave.
- Jack says: Piggy is lying.
- Jack says: Piggy and I are the same type.